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Variations in Home Advantage: Evidence
from the National Hockey League

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Abstract

We study two important aspects of the home advantage for ice hockey. First we study team and season level data to uncover systematic variation in the home advantage across teams and over time. We find no systematic variation in home advantage across teams. Secondly, we use six seasons of game-level data to uncover sources of the home advantage. Our results show that additional rest for the home team has no impact on the win-loss probability. We find evidence that the home team's rule advantage with regard to stick placement during face-offs helps to explain the home advantage, but that the home team's rule advantage with regard to last line change opportunity does not. We find that the home advantage varies over the progression of the season, being the highest at the beginning and end of the season, and very small in the middle of the season. We also find that the home advantage is larger at the end of the season when the two teams are high in the standings. Some of these results are consistent with referee bias, but we are unable to show a causal relationship between referee bias and these results.

KEYWORDS: home advantage, game-level data

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1. INTRODUCTION

A well-known feature of sporting contests is the home-team advantage whereby the home team consistently wins more frequently than its opponent. This is a natural outcome that benefits team owners since it promotes higher attendance and ticket sales. All major team sports leagues guarantee each team play a balanced schedule where half of all games are played at home and half away, so that no team has a potential for a larger home advantage over the season. In spite of this balance, the home advantage varies widely across teams and over time. During the past six seasons of the National Hockey League (NHL), the Boston Bruins had the lowest average home advantage of 2%, with a 50% win percentage at home and a 48% win percentage on the road. In the 2010/11 season during which the Bruins won the Stanley cup, their winning percentage during regular season play was better on the road (59%) than playing at home (54%), giving it a home *dis*advantage of 5%. On the other hand, the Calgary Flames in the 2006/07 season had a substantially larger win percentage at home (73%) than on the road (30%). Even though the home-team advantage is an empirical regularity, each sporting contest is a random event, so why is it that the home team wins more frequently? In this paper, we study two different aspects of the home advantage in ice hockey. First, we ask whether the size of the home advantage varies systematically across teams and over time, by analyzing season-level data for all teams in the National Hockey League. The 2010/11 season showed a historically small average home advantage of less than 4% whereas the prior 2009/10 season showed a rather large average advantage of over 12%. Looking at team-specific data shows even more variation. In this paper we seek to uncover persistent differences across teams and over time in the size of the advantage. Secondly, we use game-level data to estimate win-loss probability models in an effort to uncover specific sources of the advantage. A team's performance, at home and on the road, reflects many aspects of play other than the home ice advantage. Although the existence of a home advantage is undisputed, there remains much speculation as to its sources. Psychological factors including more rest for the home team, crowd support, referee bias and rules advantages are typically cited as sources of the home advantage.

We use six seasons of game-level data for the NHL to examine how the home ice advantage varies across teams and over time. We find that there is no systematic pattern to the home advantage. It does not appear to differ significantly across teams, although it does show variation across seasons. With regard to the sources of the home advantage, we find no evidence that rest between games has any effect on the home advantage. We find that the home team's rule advantage pertaining to face-offs is persistent across all six seasons of data, but that it does not explain all of the home advantage. Furthermore, the

home team's rule advantage concerning the last line change opportunity does not explain any of the home advantage. We analyze the effect of attendance on the likelihood that the home team wins and find that attendance has an ambiguous impact on the likelihood of winning. Lastly, we conduct a number of analyses suggested by the literature on referee bias. While we do not claim to find solid evidence of referee bias, we do find results that are consistent with it. We separate games into games that end in regulation time with a one goal differential, those that end in overtime and those that end in a shootout. We find that the size of the home advantage is larger for games that end in overtime than for those that end in regulation time with a one goal differential, yet we find there to be no home advantage in games that end in shootouts. We also find that the size of the home advantage varies as the season progresses, with it being highest early and late in the season and nearly nonexistent in the middle of the season. Lastly, we find a larger home advantage when the two teams are both high in standings near the end of the season where there is a lot at stake. A review of the literature appears in Section 2. A discussion of the data and descriptive statistics on the size of the home advantage appear in Section 3. Analysis of team-varying and time-varying home advantage is presented in Section 4. Sources of the home advantage are analyzed with game-level data in Section 5. Section 6 concludes.

2. REVIEW OF THE LITERATURE

Research into variations of the home advantage across teams and over time is not as plentiful as the research into the sources of the home advantage. Harville and Smith (1994) find team-by-team variability in home advantage in college basketball to be relatively small. This result was reinforced for professional basketball by Jones (2008) who finds team-specific home advantages in the National Basketball Association to lack statistical reliability. On the other hand, Clarke (2005) finds that home-team advantage varies systematically across teams in Australian football and believes it is due to differences in crowd effects. Jamieson (2010) addresses many aspects of home advantage, including variation over time. He suggests that the home advantage can be expected to rise with the recent increase in media coverage, which exploded with the growth of telecommunications. However, he also suggests that the recent increase in turnover among the players on a team can diminish the size of the home advantage. His only conclusion is that the home advantage was typically much larger in most sports prior to 1950.

Although the presence of a home advantage is well established, unraveling the effect of the home crowd has been difficult, which in turn has resulted in a large body of research into the phenomenon. Courneya and Carron (1992)

provide an extensive review of the literature, which was updated by Carron, Loughhead, and Bray (2005), and many researchers have drawn upon these reviews. Popular explanations include more rest and more favorable rules for the home team and crowd support. Entine and Small (2008) study the effect of rest on the home advantage for basketball, and find that more rest for the home team contributes to the home-court advantage. Leard and Doyle (2011) use a single season of data for the National Hockey League (NHL), and find that none of the home advantage could be explained by the home team having more rest between games, but that a rule advantage for the home team with regard to face-off stick placements explained part of the advantage. Although these results were fairly strong, they pertain to a single season of data. The impact of rules advantages for the home team have been studied in a number of sports, yet large differences in rules across sports makes a summary of the work difficult. As for hockey, Liardi and Carron (2011) use pairwise correlations and find no significant correlation between face-off success for the home team and winning.

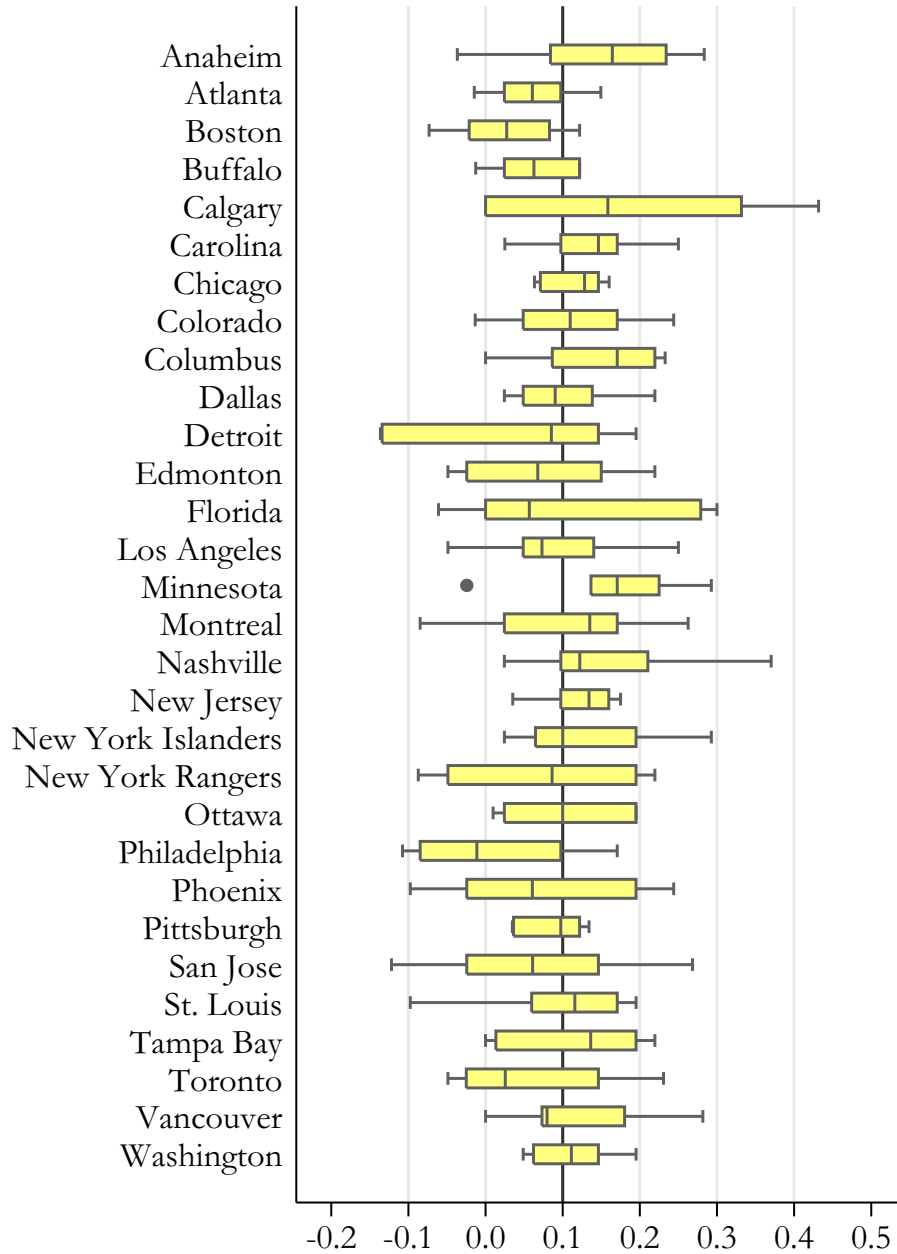
In their book *Scorecasting*, Moskowitz and Wertheim (2011) argue that most home-team advantage can be traced to referee bias. They rule out the effect of the crowd by showing that the home and away teams are equally likely to score during shootouts in hockey and free-throw percentages in basketball. The fact that the home team doesn't perform any better than the away team in these situations, they argue, is evidence against the influence of the home crowd. However, the home crowd can have an indirect effect through the behavior of the referees. Pettersson-Lidbom and Priks (2010) find that referees punish away teams more harshly than home teams when games are played in front of fans compared to games played in empty arenas. Garicano et al. (2005) find evidence of referee bias in European soccer, but that it was dependent on the closeness of the game. In close games, referees showed a larger bias in favor of the home team by adding more minutes for injury time when the home team was behind by one goal, but not when the home team was behind by more than one goal, the extra time allowing the home team a better chance to score.

3. DATA AND DESCRIPTIVE STATISTICS

Game-level data for six seasons of regular season play in the National Hockey League (NHL) from 2005/06 – 2010/11 are analyzed.¹ A team's home advantage for a season is the difference between its win ratio at home and away.

¹ The majority of our data came from the National Hockey League's web site www.nhl.com. Data on face-offs won and data used to construct the CORSI variable came from Gabriel Desjardins at Behindthenet.ca.

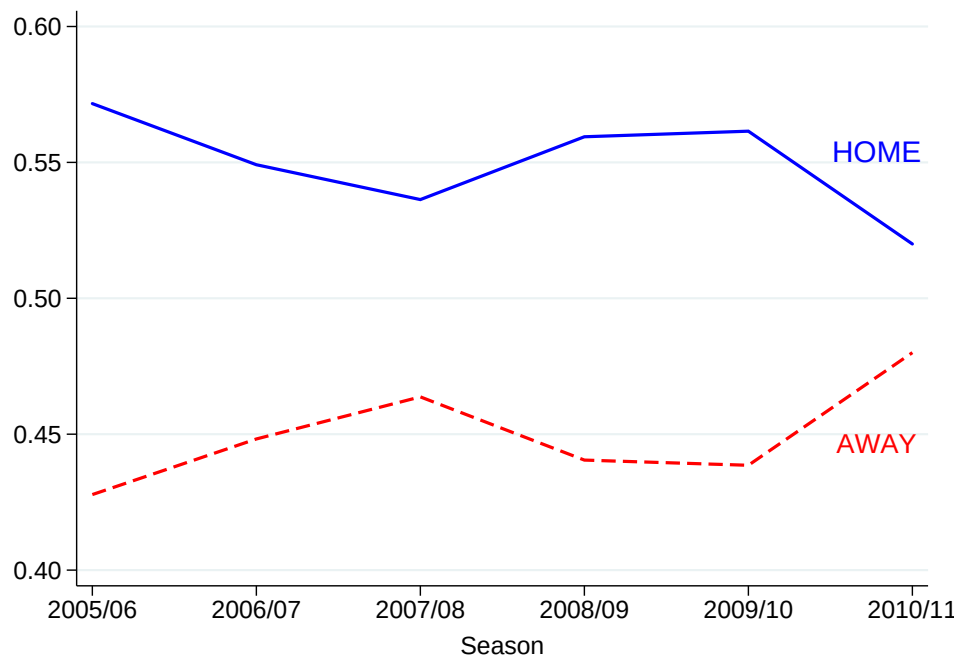
Figure 1A: Home Advantage by Team 2005/06 -2010/11
Difference Between Home and Away Win Ratios²



² We use the win ratio, which lies in $[0,1]$, instead of a win percentage (%), to facilitate comparison with the probit models in Section 5.

The distributions of these ratios over the six seasons of data for each team appear in Figure 1A, with the overall mean being 0.10 (10% home advantage). Calgary had the largest home advantage of 0.43 (43%) in the 2006/07 season when it won 73% of its home games but only 30% of its away games. The absence of a home advantage is not unusual. In the 2005/06 season, Detroit logged the worse differential of -0.14 (-14%) when it won 66% of its home games but 80% of its away games. Figure 1B shows the trends in the average team win ratio at home and on the road. The average home advantage was largest at the start of our sample and lowest at the end, with seasonal fluctuations in between and no clear trend.

Figure 1B: Average Team Home and Away Win Ratios over Time



Differences in performance statistics for the home and away teams appear in Table 2. We find that goalie performance, measured as goalie save-percentage *GSVPCT*, shows the home goalie averages 90.4% and the away goalie, 90.1%. The home team, on average, takes two more shots on goal (*SHOG*) than the away team. This may be explained to some extent by recognizing that the home team wins more than two additional face-offs (*FACE*) compared to the away team. Puck possession is measured with the Corsi variable (*CORSI*), which is the sum of shots directed at the net. These shots include shots on goal as well as blocked

shots and missed shots and are strongly correlated with puck possessions in the offensive zone.³ We then measure the efficiency of the team's puck possessions with *POEFF*, which is the ratio of shots on goal to *CORSI* in %. Of the 7,380 games played over these six seasons, we eliminated 81 games due to suspicious data for *CORSI* and 2 due to unavailable data on face-offs won. We find that the home team, on average, has a significantly larger Corsi measure, but that these additional puck possessions are not accompanied by greater efficiency since *POEFF* is not statistically different for the home and away teams. The average home *POEFF* is 55.11%, meaning that on average a team gets about one shot on goal for every two offensive zone possessions. We find that the away team receives roughly 45 seconds more in penalty minutes (*PIM*), resulting in the home team having about three-eighths of an additional power play opportunity (*PPLAY*). Total face-offs (*TFACE*) is the sum of the face offs won by the home team and those won by the away team. On average a game has about 58 face-offs. In addition to the variables presented in Table 2, we also present data in Table 4 on the number of days off (*DAYSSOFF*) a team has had since its last game. We find that the home team has had slightly more time off than the away team.

Table 2: Performance Measures 2005/06 – 2010/11

Variable	HOME			AWAY			t-stat	p-value
	N	Mean	SD	N	Mean	SD		
WIN (%)	7380	55	0.50	7380	45	0.50	12.21	< 0.001
GSVPCT (%)	7380	90.43	6.05	7380	90.08	5.87	3.53	< 0.001
SHOG	7380	30.94	6.81	7380	28.90	6.47	18.63	< 0.001
CORSI	7297	56.39	10.58	7297	53.00	10.43	19.52	< 0.001
POEFF (%)	7297	55.11	8.08	7297	54.87	8.36	1.77	0.077
FACE	7378	29.98	5.71	7378	27.61	5.56	25.48	< 0.001
PPLAY	7380	4.58	2.08	7380	4.22	1.98	10.80	< 0.001
PIM	7380	13.29	8.94	7380	14.08	9.30	-5.25	< 0.001
TFACE	7378	57.59	7.63					

There are 30 teams that play 82 games per season, 41 games at home and 41 games on the road, for a total of 1230 games per season. With 6 seasons of data, we have 7380 observations. For our model estimations, we removed 81 games with suspicious data on *CORSI* and 2 games due to missing data on face-offs. We also eliminated 5 games where the home team played in a venue that is not its home ice (e.g. when Boston Bruins played hockey in Fenway Park on Jan 1, 2010).

³ See Desgardins (n.d.) for the development of this measure of puck possession, which is credited to Jim Corsi, the goaltending coach for the Buffalo Sabres and Vic Ferrari, from the Edmonton Oilers.

4. HOME ADVANTAGE ACROSS TEAMS AND OVER TIME

In this section we estimate models of home ice advantage using a panel of season-level data to determine if the advantage varies systematically across teams and over time. We have six seasons of data for each of the 30 teams. These 180 observations are used to estimate models of home ice advantage as the dependent variable. For each team i we construct its win ratio for season t when playing at home ($WINH_{it}$) and when playing away ($WINA_{it}$), and then difference these ratios to arrive at the home advantage, similar to Bray (1999), for a dependent variable $DWIN_{it} = WINH_{it} - WINA_{it}$. We estimate a set of models of the form:

$$(1) \quad DWIN_{it} = \beta_o + \sum_j \beta_j X_{jit} + \alpha_i + \lambda_t + u_{it}$$

The explanatory variables X_{jit} are chosen to capture the differences between the performances of the teams at home and on the road. These include differences in home and away goalie save percentage ($DGSPCT$), power play opportunities ($DPPLAY$), possession efficiency ($DPOEFF$), and face-offs won ($DFACE$). The term α_i is an unobserved team effect that varies across teams but does not vary over time. A team effect would include such factors as hockey tradition, stadium design, and demographic differences in preferences for hockey. One might argue that these factors do vary over time, but any variation over the six seasons of data is likely to be small so we can reasonably assume them to be constant over time. The term λ_t represents unobserved time effects and is included to capture variation in home advantage that is constant across teams, but varies over time. These unobserved effects would include changes to game rules and other league wide shocks that might impact the home-team advantage. The error term u_{it} captures unmeasured variation in home advantage that varies over teams and seasons, and it is assumed that $u_{it} \sim \text{iid}(0, \sigma_u)$. We estimate (1) by starting with a model that has only a constant term, and then sequentially build up to a model that includes all factors expressed in (1). How we estimate (1) depends on the assumptions we make about α_i and λ_t . Together these factors represent unobserved heterogeneity that, if ignored, could lead to inconsistent estimation of the β parameters. A fixed effects specification allows $E(\alpha_i|X_{it})$ and $E(\lambda_t|X_{it})$ to vary over X_{it} , meaning the effects can be correlated with the observed explanatory variables. These effects are incorporated into the model by including, as independent variables, a set of dummy variables for each set of effects and applying ordinary least squares. We then test the two groups of dummy variables for joint statistical significance with an F test. If a group is statistically significant, then the data suggest that there are meaningful differences in home advantages across the group. An alternative estimation strategy is to assume that the unobserved effects α_i and λ_t are uncorrelated with the regressors. Specifically, it is assumed that $\alpha_i \sim \text{iid}(\alpha, \sigma_\alpha)$ and

$\lambda_t \sim \text{iid}(\lambda, \sigma_\lambda)$, and these random variables are incorporated into the model's error term. This random effects model is estimated using feasible generalized least squares, which will estimate β_j , σ_α , σ_λ , and σ_u . The parameters α and λ are subsumed into the model's intercept, β_o , and cannot be separately identified. This random effects model is more efficient than fixed effects if the assumption of uncorrelatedness is supported. We estimate (1) both ways and conduct a Hausman test to choose between fixed and random effects.⁴ The results appear in Table 3.

Table 3: Home Ice Advantage (*DWIN*)
Team-Season Data 2005/06 – 2010/11

VARIABLES	(1) OLS	(2) OLS ^a	(3) Fixed Effects	(4) Random Effects
Constant	0.100** (0.008)	0.068** (0.014)	--	0.072** (0.019)
DGSVPCT		0.036** (0.005)	0.038** (0.006)	0.036** (0.005)
DPOEFF		-0.001 (0.002)	-0.002 (0.004)	-0.001 (0.002)
DPPLAY		0.0005 (0.0004)	0.0005 (0.0004)	0.0005 (0.0004)
DFACE		0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
σ_α				0.016
σ_λ				0.033
F-stat: α_i			1.19 [0.25]	
F-stat: λ_i			5.05 [<0.001]	
Hausman Test $\chi^2(4)$			0.80 [0.94]	
Breusch-Pagan $\chi^2(2)$				24.59 [<0.001]
R ²	--	0.206	0.444	0.312
adj R ²	--	0.188	0.295	0.296
N	180	180	180	180

** p<0.01, * p<0.05, + p<0.1. Standard errors in parentheses. *p*-values in brackets.

^aThe residuals from the OLS model in column 2 were tested for group-wise heteroskedasticity. The null hypothesis of a homoskedastic error could not be rejected, for team groups and season groups. The *p*-values for the *F*-statistics were 0.504 and 0.840.

⁴ See Cameron and Trivedi (2005) and Wooldridge (2002).

Column 1 shows a baseline home advantage of 0.10 (a 10% advantage), with no independent variables or effects included. This estimate coincides with the average home advantage over the six seasons from Table 2. In column 2 we include the set of explanatory variables that measure differences in the teams' home and away performances measures, and estimate the model using ordinary least squares with no estimation of the team or season effects. Here we find that only goalie performance is statistically significant in explaining some of the home advantage, whereby a one percentage point improvement in the home vs. away goalie save percentage predicts an increase in the home advantage of 0.036 (3.6 percentage points). The fixed effects estimation results appear in column 3 where we have included a set of fixed effects for teams and a set for the seasons. The F -statistic that tests the hypothesis that the team fixed effects are not different from each other is 1.19 with a p -value of 0.25, implying that the home advantage shows no systematic variation across teams. The season effects are statistically significant, with an F -statistic of 5.05 and a p -value less than 0.001. Finally, in column 4 we allow the effects to be random. The Hausman test compares the coefficient estimates for the fixed effects and random effects specifications and is designed to test the hypothesis that the random effects (α_i and λ_t) are uncorrelated with the X_{it} variables. If the hypothesis cannot be rejected, the random effects results are preferred over the fixed effects due to improved efficiency. In our results, the Hausman test fails to reject the null. Lastly, we conduct a Breusch-Pagan test to compare the random effects to the OLS results. The null hypothesis is that there are no random effects, which we are able to reject, although this rejection is being driven by variation in the season effects. The estimate of σ_λ is 0.033 (3.3%), which we interpret as the standard deviation in the home advantage over the seasons, controlling for the explanatory variables. The estimate of σ_α , the standard deviation in home advantage over the teams, is only half as large at 0.016 (1.6%). We have established that the home advantage does not vary systematically across teams, although it does vary over time and this time variation is best captured by random effects and not by fixed effects.

5. HOME ADVANTAGE FROM GAME-LEVEL DATA

5.1 INTRODUCTION

The problem with using season-level data is that some of the more interesting hypotheses behind the home advantage cannot be adequately tested. Specifically, the hypothesis that the home team benefits from more rest than the away team and has an advantage with regards to rules such as stick placement during face-offs and line changes during play should be tested using game-level data. Table 4

shows a frequency distribution for the number of days off for the home team and the away team for all games in the six seasons of data. We find more than twice as many occurrences of the away team playing back-to-back games with no days off compared to the home team. 88.3% of the time the home team has had at least one day of rest, but only 74.2% of the time has the away team had at least one day of rest. In the NHL, the home team enjoys rules advantages with regard to face-offs and line changes. We saw in Table 2 that the home team wins more face-offs than the away team, and some of this difference can be attributed to the fact that the home team player places his stick last during a face-off, which is commonly believed to be advantageous for winning the face-off. Furthermore, the home team has the last line change opportunity, allowing the coach to alter the line of players without the away team coach being able to respond. Lastly, referees may be biased in favor of the home team and this bias may work through the crowd.

Table 4:
Days Off Since Last Game 2005/06 – 2010/11

DAYSOFF	Home Team		Away Team	
	Frequency	%	Frequency	%
0	852	11.7	1885	25.8
1	4136	56.8	3450	47.3
2	1485	20.4	1356	18.6
3	496	6.8	370	5.1
4	190	2.6	121	1.7
> 4	126	1.7	113	1.6
# of Games	7285		7295	
Average	1.42		1.19	

DAYSOFF is measured for each team starting with their second game of the season so that each team will have a missing observation for DAYSOFF on their first game of the season. Close to half, although not exactly half, of all teams play their first game at home, which means that the number of observations for DAYSOFF for the home and away teams will be close but not identical.

5.2 MODEL OF WIN-LOSS PROBABILITY

Using game-level data, we can test for several sources of home advantage mentioned above. To do so, we structure our game-level data in such a way that allows us to incorporate the abilities of both teams by measuring the performance variables as differentials. This data organization was used by Carmichael et al. (2000) and Leard and Doyle (2011). Since the game is the unit of observation,

and each game has a home team and an away team, we randomly choose one of the two teams to be the “observed” team and the other the “opponent”, providing us with a set of observed teams that is independent. We then model the probability that the observed team wins the game. Using this data structure allows each game to appear in the data only once, and it facilitates the identification of a home-ice advantage since the home team can be identified with a dummy variable. The probability of winning a game is estimated using a probit model. The dichotomous dependent variable WIN_{it} equals 1 if the observed team i wins game t and it equals 0 if it loses. This binomial random variable has a probability of success, denoted π_{it} , that is modeled as a function of covariates. The probit model uses the cumulative normal distribution function $\Phi(\cdot)$ to model the probability of success:

$$WIN_{it} \sim BINOMIAL(\pi_{it})$$

$$(2) \quad \pi_{it} = P(WIN_{it} = 1) = \Phi\left(\beta_o + \sum_j \beta_j X_{jit}\right)$$

Using data on the covariates X_{jit} and actual game outcomes WIN_{it} , we estimate the parameters β_j using maximum likelihood estimation. The predicted probability of winning is calculated as:

$$(3) \quad \hat{\pi}_{it} = \Phi\left(\hat{\beta}_o + \sum_j \hat{\beta}_j X_{jit}\right)$$

These probabilities will lie in the unit interval $[0,1]$ and vary over i and t . Unlike a linear regression model, the impact of the covariates on the probability of winning is not a simple constant for the probit model. We have the following equation for the marginal effect of X_j on the predicted probability of winning, which results from differentiating (3) with respect to X_j :

$$(4) \quad \frac{\partial \hat{\pi}_{it}}{\partial X_{jit}} = \phi\left(\hat{\beta}_o + \sum_j \hat{\beta}_j X_{jit}\right) \hat{\beta}_j$$

$\phi(\cdot)$ is the normal density function. Since $\phi(\cdot)$ is a positive valued function, the sign of $\hat{\beta}_j$ tells us the direction of the marginal impact, but in general, the marginal effect will differ across the sample as the values of the covariates differ.

We follow standard procedure and report average marginal effects, which involves evaluating (4) at each observation and reporting the average.⁵

We estimate the size of the home advantage by including in the set of covariates a dummy variable, *HOME*, that equals 1 if the observed team is the home team, and it equals 0 if the observed team is the away team. We estimate the model and sequentially add independent variables to examine their impact on the *HOME* coefficient. We first add differential days off (*DAYSOFF*). If common thinking is correct, the coefficient on *DAYSOFF* should be positive, meaning that if the observed team has more days off than the opponent, regardless of which team is playing at home, there is a higher probability of winning. Furthermore, if rest is a primary source behind the home advantage, then we would expect to see the coefficient on *HOME* reduced with the inclusion of *DAYSOFF*. We also include our measures of defensive and offensive ability. These are the differential of goalie save percentage (*GSVPCT*) to capture relative defensive ability between the two teams and differential possession efficiency (*POEFF*) and differential face-offs won (*FACE*) for our offensive variables. To the extent that the home team's rule advantage with respect to face-offs benefits the home team, we would expect to see the coefficient on *HOME* diminish when we include *FACE*. We include differential power play opportunities (*PPLAY*) to capture differences in the number of times the team has more players on the ice than the opponent. Lastly, we include a variable that measures the difference in the number of points won to date per game played in the season for the two teams (*LPTSG*). This variable is included to capture relative team qualities that the set of differential performance variables might miss. It is similar to a measure of team standings, but instead of differential total points earned to date, which might vary between the two teams because they have played a different number of games to date and not because of different abilities, we measure differential points earned relative to the number of games played prior to the current game. We expect it to have a positive impact on the probability of winning.

The results appear in Table 5. We find that the home advantage, when calculated as a marginal effect from the probit results, is initially estimated at 0.10 (10%) prior to including any covariates. This home advantage coincides with the size of the home advantage from season-level data in Figure 1A. For the second model in Table 5, we include *DAYSOFF* and find absolutely no impact on the home advantage with the inclusion of differential days off, since the coefficient on *HOME* changes very little. Not only does the inclusion of *DAYSOFF* have no

⁵ If X_j is a dummy variable, marginal effects are not calculated by differentiating (3), but instead are found by evaluating (3) twice: once at $X=1$ and again at $X=0$ and the other variables at their sample means. The difference in these predicted probabilities is the marginal effect of the dummy variable.

effect on the *HOME* dummy, its coefficient is statistically insignificant.⁶ The inclusion of differential goalie performance, possession efficiency, and power play opportunities in the third model of Table 5 has a sizable impact on the home advantage, reducing it to 0.039 (3.9%). This marginal effect for the *HOME* variable can be thought of as “unaccounted for” home advantage. The impact of differential days off remains insignificant. This result is not sensitive to the order in which *DAYSOFF* is entered into the model. In the last model in Table 5, we have included differential face-offs won and find it to have an impact on the *HOME* coefficient and marginal effect, with the “unaccounted for” home advantage falling to 0.03 (3%), meaning that if the away team were to perform identically to the home team with regard to the offensive and defensive performance variables, the home team is still left with a 3% advantage.

The marginal effects in Table 5 measure the impact of the explanatory variables on the predicted probability of the observed team winning the game. Since the predicted probabilities lie in the range $[0,1]$, the marginal effects measure change in this range, and are best understood by considering each (X_{it}) variable’s unit of measurement and a sensible change (ΔX_{it}) for analysis. For *GSVPCT*, we find that an increase in the observed team’s goalie save percentage of one point over its opponent is predicted to increase the probability of winning by 0.04, or 4 percentage points, holding the other variables constant. A 5 percentage point goalie save percentage will increase the probability by 20 percentage points. An increase in puck efficiency (*POEFF*) of 1 percentage point better than its opponent is predicted to increase the probability of winning by 0.007, or 0.7 of a percentage point, and by 7 percentage points when puck efficiency increases by 10 percentage points over its opponent. One power play opportunity more than the opponent is predicted to increase the probability of winning by 1.6 percentage points, holding the other factors constant. The variable points earned per game (*LPTS*) is statistically significant. For interpretation of its marginal effect, it is useful to know that the average points per game lies between 0 and 2 and has an average of about 1.11 with a standard deviation of 0.26.⁷ So a team that has earned a half of a point more per game compared to its opponent ($\Delta LPTS = 0.5$) is predicted to have a probability of winning that is 2.7 percentage points higher ($0.054 \times 0.5 = 0.027$), holding other factors constant. The marginal effect of face-offs won on the probability of winning is 0.002, meaning that for every additional face-off won compared to its opponent, a team’s probability of winning increases by 0.2 of a percentage point, holding the other factors constant, so a 10 face-off won differential increases the probability by 2

⁶ We estimated the probit model with an interaction term between *HOME* and *DAYSOFF*. The interaction term was statistically insignificant with a p -value of 0.26.

⁷ Because games that end in regulation time “pay out” 2 points but games that go to overtime “pay out” 3 points, the average of points earned per game will be greater than 1.

Table 5: Home Advantage: Days Off and Face-Off Rule Advantage
Game Data NHL 2005/06 – 2010/11

Variable	(1) $\hat{\beta}$ Marg. Effect	(2) $\hat{\beta}$ Marg. Effect	(3) $\hat{\beta}$ Marg. Effect	(4) $\hat{\beta}$ Marg. Effect
HOME	0.251** (0.030)	0.247** (0.030)	0.307** (0.051)	0.234** (0.054)
DAYSOFF		0.010 (0.011)	-0.001 (0.018)	-0.003 (0.018)
GSVPCT			0.313** (0.010)	0.316** (0.010)
POEFF			0.051** (0.003)	0.052** (0.003)
PPLAY			0.129** (0.012)	0.129** (0.012)
LPTSG			0.455** (0.067)	0.426** (0.067)
FACE				0.016** (0.003)
N	7,190	7,190	7,188	7,188
Ln(L)	-4948	-4947	-1621	-1607
count R ²	0.550	0.550	0.903	0.904
adj count R ²	0.0961	0.0964	0.805	0.807

Robust standard errors in parentheses.

** p<0.01, * p<0.05, + p<0.1 Except for HOME, all variables are measured as differentials between the observed team and its opponent. Except for HOME, the marginal effects are calculated evaluating equation (5) at each observation, and averaging over the observations. Because the predicted probabilities lie in the range [0,1], the marginal effects measure change in this range.

percentage points. The results in Table 5 suggest that only one percentage point of the home advantage can be attributed to the face-off rules advantage afforded the home team and that absolutely none of it can be attributed to more rest for the home team. We might be tempted to conclude that six-tenths (6 percentage points of the 10% home advantage) can be attributed to the joint effect of goalie performance, possession efficiency and power play opportunities, leaving the “unaccounted for” home advantage at 0.03 (3%). However, the effects of these differential performance variables do not provide an explanation of the home advantage, since we have yet to explain why the home team performs better than the away team in these measures. Regarding the face-off variable, we have an explanation for its significance and its impact on the *HOME* dummy, namely the face-off rule advantage granted to the home team.

5.3 EFFECT OF ATTENDANCE

The effect of the crowd as a contributing factor behind the home advantage was dismissed by Moskowitz and Wertheim (2011) by citing the observation that the performance of the home team is no better than the away team for free throws in basketball and for shootouts in hockey. This conclusion is not necessarily enough to debunk the hypothesis of the influence of the crowd, since a free throw in basketball involves pure shooting ability and no defensive ability on the part of the opposing team. As for hockey, a shootout occurs after the two teams have played 60 minutes of regulation play and 5 minutes of overtime play. In this case, the shootout only captures the ability of the goalie versus a single offensive player. In fact, Smith and Groetzinger (2010) find a positive impact of attendance on the home-team advantage in baseball. We test for the effect of attendance on the probability that the *home* team wins by reorganizing our data such that the dependent variable is a dichotomous variable indicating whether the home team wins or loses. The independent variables are similar to those in Table 5 except that the performance differentials are constructed as the difference between the home and the away teams’ performance. We exclude the home dummy, which is no longer needed, and include an attendance variable (*PERATT*) that measures attendance as a percentage of capacity. We hypothesize that a larger crowd bodes well for the home team, and we expect a positive coefficient on *PERATT*. Of course, it could be that the crowd effect is not adequately measured by *PERATT*, but given our data it is our only measure for the crowd.

This win-loss model for the home team can be used to pick up the effect of the home team’s last line change rule advantage. As discussed earlier, the home team is granted rules advantages in the NHL, such as the face-off stick placement

rule and also the opportunity to make the last line change before play resumes. The number of these last-line change opportunities aligns with the number of play stoppages, because the last line change opportunity occurs when play resumes. We use the total number of face-offs (*TFACE*) in the game as a proxy for the number of these last line change opportunities, since the resumption of play begins with a face-off.⁸ If the last line change opportunity is beneficial to the home team, we would expect that more opportunities for changing the line would increase the probability of winning, holding the other covariates constant.

The win-loss probability model with attendance as an explanatory likely suffers from endogeneity bias as discussed by Smith and Groetzinger (2010), whereby attendance not only may improve the chances of the home team winning, but also a higher expectation of a win will attract a larger crowd. That is, home teams who are expected to win will attract a larger crowd so a positive coefficient on attendance in the win-loss probit model cannot be interpreted as evidence that the crowd improves the probability of winning. We replicate their solution whereby we use instrumental variables estimation to mitigate the endogeneity bias. For instruments, we use the day of the week and a set of team effects. These results appear in Table 6, with the first model containing results using only attendance as a percent of capacity and differential days off. We then add the differential performance variables that were used in the previous probit models. In column 1 we find that attendance has a small but significant positive effect on the probability of winning although differential days off does not. When we add the differential performance variables, the impact of attendance becomes statistically insignificant. The *PERATT* variable is highly skewed, with several teams at 100% capacity for all games. It could well be that *PERATT* doesn't provide enough variation after the endogenous variation is removed by the instrumental variables. Lastly, we find that total face-offs, our proxy for the home team's advantage of having the last line change opportunity before the resumption of play, has no measureable impact on the probability that the home team wins the game. It should be noted that the coefficient estimates in model 2 of Table 6 are nearly identical to those for model 4 in Table 5, where we had randomized the team whose win-loss probability is modeled. The fact that these two sets of coefficient estimates are quite close confirms the claim that the randomization of the teams into observed and opponent teams for Table 5 does not skew the results.

⁸ The exception is any team that "ices" the puck will force a stoppage and the offending team is not allowed a line change. We do not have data on icing for our full set of games so we cannot remove instances where the home team ices the puck and loses its ability to change the line. This means that the measure of total face-offs will overstate the number of line change opportunities. However, we took a random sample of 30 games and measured the incidences of icing. We found that the average number of icings is about 5 and that the difference in icings between the home team and the away team is not statistically significant.

Table 6: Instrumental Variables Probit Results
Game Data NHL 2005/06 – 2010/11

VARIABLES	(1)		(2)		(3)	
	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>
PERATT	0.006** (0.002)	0.002	-0.003 (0.003)	-0.0004	-0.003 (0.003)	-0.0004
DAYSOFF	0.009 (0.011)	0.005	-0.004 (0.018)	-0.001	-0.004 (0.018)	-0.001
GSVPCT			0.315** (0.010)	0.040	0.315** (0.010)	0.040
POEFF			0.052** (0.003)	0.007	0.052** (0.003)	0.007
PPLAY			0.128** (0.012)	0.016	0.128** (0.012)	0.016
LPTSG			0.436** (0.068)	0.055	0.436** (0.068)	0.055
FACE			0.016** (0.003)	0.002	0.016** (0.003)	0.002
TFACE					0.002 (0.003)	0.0004
N	7,190		7,188		7,188	
count R ²	0.549		0.904		0.903	
adj count R ²	0.505		0.826		0.824	

** p<0.01, * p<0.05, + p<0.1.

Robust standard errors in parentheses. Except for PERATT, all variables are measured as differentials between the home team and the away team. The marginal effects (*m.e.*) are calculated by evaluating equation (5) at each observation, and averaging over the observations. Because the predicted probabilities lie in the range [0,1], the marginal effects measure change in this range.

5.4 LOOKING FOR REFEREE BIAS

In studies of soccer and baseball, researchers have analyzed referee bias and have argued that the referees balance the costs against the benefits of making biased calls in favor of the home team. The hypothesis is that a referee is more likely to side with the home team when the game is close than when the score differential

is large. To look for this effect in the hockey data, we want to identify games that are close and measure the size of home advantage. Our problem is that we only have end-of-game data, yet the closeness of a game will vary during play. Our approach is to take the sample of games that go into overtime as well as those games that end in regulation time with a one goal differential to create our sample of close games. We examine the rate at which the home team wins for three types of games: those games that end in regulation time with a one goal differential (REGTIME), those games that end in overtime with a goal scored (OT-REG) and those games that end in overtime with a shootout (OT-SO). Table 7A shows the frequency of overtime games and shootouts in particular as well as the size of the home advantage in shootouts. Desjardins (2009) suggested that the adoption of the shootout rule in the 2005-06 season lead to an increase in the number of games going into overtime and then a shootout since there appears to be even chances of winning the shootout. The statistics in Table 7A suggest that to be the case, where the home advantage in shootouts is fairly nonexistent, and is quite negative in the last season, with the away team winning more than 60% of the shootouts.

Table 7A: Overtime and Shootout Frequencies
Game Data NHL 2005/06 – 2010/11

Season	% of Games : Overtime	% of Overtime Games going to Shootout	Home Advantage in Shootouts
2005-06	22.9%	51.5%	5.7%
2006-07	22.2%	60.2%	-4.5%
2007-08	22.1%	57.3%	-2.0%
2008-09	22.9%	57.0%	-3.8%
2009-10	24.3%	61.2%	1.1%
2010-11	24.2%	50.7%	-21.1%

The top row of Table 7B contains the home advantage for these three subsets of games averaged over all six seasons. The average home advantage for “close” regulation time games was 5.5%, it was 6.3% for games that ended in overtime with an overtime goal, but it was negative 4% for games that ended in a shootout. We also estimate our win-loss probability model using these three subsets of games. The three subsets of games provide very different results, especially with respect to the estimate of the home advantage. Close games that end in regulation time have a home advantage of 5.5% and the differential offensive and defensive variables account for roughly two of these percentage points, leaving an “unexplained” home advantage of 3.8%. For games that go into overtime, the off-

Table 7B: Home Advantage: Close Games and Overtime Games
Game Data NHL 2005/06 – 2010/11

	Regulation Games		Overtime Games			
	Goal Difference = 1		REGULAR OT		SHOOTOUT	
Avg. Home Advantage	0.055 (5.5%)		0.063 (6.3%)		-0.04 (-4.0%)	
Variable	$\hat{\beta}$	<i>Marginal Effect</i>	$\hat{\beta}$	<i>Marginal Effect</i>	$\hat{\beta}$	<i>Marginal Effect</i>
HOME	0.303** (0.109)	0.038	0.410** (0.144)	0.083	-0.073 (0.087)	-0.029
DAYSOFF	-0.021 (0.043)	-0.003	-0.046 (0.046)	-0.009	-0.003 (0.033)	-0.001
GSVPCT	0.451** (0.032)	0.056	0.358** (0.037)	0.073	-0.016 (0.017)	-0.006
POEFF	0.080** (0.006)	0.010	0.066** (0.008)	0.013	-0.004 (0.005)	-0.002
PPLAY	0.194** (0.028)	0.024	0.218** (0.031)	0.044	-0.008 (0.019)	-0.003
LPTSG	0.562** (0.132)	0.070	0.301 ⁺ (0.178)	0.061	0.126 (0.114)	0.050
FACE	0.023** (0.007)	0.003	0.032** (0.008)	0.006	-0.008 (0.005)	-0.003
N	1,693		726		934	
$\ln(L)$	-358.1		-255.5		-644.1	
count R ²	0.938		0.868		0.541	
adj count R ²	0.876		0.721		0.057	

Robust standard errors in parentheses.

** p<0.01, * p<0.05, ⁺ p<0.1 Except for HOME, the marginal effects are calculated evaluating equation (5) at each observation, and averaging over the observations. Because the predicted probabilities lie in the range [0,1], the marginal effects measure change in this range.

ensive and defensive variables are unable to explain any of the overtime home advantage because once we include the differential variables, the home advantage actually increases to 8.3% from the raw average of 6.3%. The large difference between the regulation games and overtime games isn't that surprising when one considers that the offensive and defensive performance variables are measured over the entire course of the game, yet a game that goes into overtime will be decided by performance only in the overtime period. We do not have performance data by period of play so performance variables used for the

overtime games measure the differential performances of the two teams over the entire game. However, this problem is the same for games that end in a shootout, yet the differences in results for these two types of overtime games are quite striking. In Table 5, we saw that the home advantage was partly “explained” by a better home goalie performance, but now we find that the home goalie doesn’t perform any better than the away goalie in a shootout, so our earlier result that showed the home goalie outperforming the away goalie does not provide us with an explanation of the home advantage. Better home goalie performance could be due to referee bias, and once a hockey game progresses to a shootout, there is very little the referee can do to give the home team an edge. However, without some clever natural experiment such as the empty-stadium soccer games of Pettersson-Lidbom and Priks (2010), we cannot conclude that this result points to referee bias although the suggestion is there.

Given the fairly substantial differences in the results for the model estimated on the three subsets of games in Table 7B, we experimented with splitting the sample along other lines. We wanted to determine if the size of the home advantage varies over the season. We divided the data into quintiles based on game number, where the first quintile comprises the first 20% of the games in a season and so forth. One might surmise that the extent of referee bias would grow over the season as more is at stake with regard to the playoffs. These results appear in Table 8. We find that the home advantage is larger for the first and last quintiles. For the first and last quintiles, the unexplained home advantage is 4.9% and 5.9%, respectively, whereas the home advantage for the middle quintile is not statistically significant. This finding is interesting although it does not provide us with any evidence as to the cause of it. It could be that the crowd effect is larger at the start of the season when there is great anticipation for the upcoming season, and also at the end of the season when the playoffs loom. However, we estimated the models of Table 6 for the quintile splits and found no different effect by quintile of relative attendance (*PERATT*) on the win-loss probability. As before, the effect of the crowd is picked up by variations in its relative size. Of course, we might surmise that some late season games have more at stake than other games with regard to the playoffs, which might impact the size of the home advantage. To investigate this hypothesis, we look to see if the size of the home advantage varies in the late season according to the teams’ point standings. Playoff spots are determined by point totals, so we split the data into samples based on point totals. We first use the sample of games in which both teams had earned at least 85 points for the season. These are games in which both teams had a lot at stake. We also created samples where only one of the two teams had at least 85 points to date, and a sample in which neither team had at least 85 points, yet the game was late in the season. These results appear in Table 9. Interestingly, we find the size of the “unexplained” home advantage for games in

Table 8: Home Advantage: By Time of Season
Game Data NHL 2005/06 – 2010/11

	quintile 1		quintile 2		quintile 3		quintile 4		quintile 5	
	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>
HOME	0.403** (0.122)	0.049	0.123 (0.118)	0.015	0.127 (0.138)	0.014	0.147 (0.118)	0.020	0.449** (0.115)	0.059
DAYSOFF	0.005 (0.037)	0.001	0.044 (0.038)	0.006	-0.058 (0.043)	-0.006	0.006 (0.032)	0.001	-0.029 (0.055)	-0.004
GSVPCT	0.313** (0.021)	0.038	0.335** (0.022)	0.042	0.333** (0.036)	0.037	0.310** (0.019)	0.041	0.315** (0.021)	0.041
POEFF	0.069** (0.007)	0.008	0.037** (0.006)	0.005	0.056** (0.008)	0.006	0.048** (0.005)	0.006	0.056** (0.006)	0.007
PPLAY	0.114** (0.025)	0.014	0.121** (0.027)	0.015	0.137** (0.032)	0.015	0.135** (0.025)	0.018	0.132** (0.026)	0.017
LPTSG	0.145 (0.088)	0.018	0.536** (0.163)	0.067	0.462* (0.203)	0.051	0.923** (0.208)	0.122	1.283** (0.225)	0.166
FACE	0.018** (0.007)	0.002	0.015* (0.006)	0.002	0.014+ (0.008)	0.002	0.011+ (0.007)	0.002	0.014* (0.007)	0.002
N	1,345		1,457		1,460		1,462		1,464	
<i>ln(L)</i>	-288.6		-323.1		-280.3		-345.0		-338.1	
count R ²	0.906		0.907		0.914		0.901		0.892	
adj count R ²	0.811		0.813		0.826		0.801		0.779	

Robust standard errors in parentheses. ** p<0.01, * p<0.05, + p<0.1. Except for HOME, the marginal effects (*m.e.*) are calculated evaluating equation (5) at each observation, and averaging over the observations. Because the predicted probabilities lie in the range [0,1], the marginal effects measure change in this range.

Table 9: Late Season Games: What is at Stake
Game Data NHL 2005/06 – 2010/11

	Both Teams at least 85 points		Observed Team has at least 85 points		Opponent has at least 85 points		Both Teams have less than 85 points	
	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>	$\hat{\beta}$	<i>m.e.</i>
HOME	1.001** (0.311)	0.092	0.320+ (0.163)	0.042	0.274 (0.236)	0.030	0.427* (0.199)	0.051
DAYSOFF	0.014 (0.114)	0.001	-0.043 (0.046)	-0.006	-0.058 (0.054)	-0.006	0.014 (0.106)	0.002
GSVPCT	0.513** (0.071)	0.047	0.304** (0.023)	0.040	0.369** (0.034)	0.040	0.325** (0.035)	0.039
POEFF	0.054** (0.016)	0.005	0.051** (0.008)	0.007	0.067** (0.014)	0.007	0.069** (0.009)	0.008
PPLAY	0.263** (0.075)	0.024	0.104** (0.036)	0.014	0.255** (0.059)	0.028	0.155** (0.043)	0.018
LPTSG	-1.594 (1.247)	-0.145	1.129** (0.245)	0.148	1.764* (0.703)	0.193	1.820** (0.493)	0.216
FACE	0.009 (0.020)	0.001	0.011 (0.010)	0.001	0.017 (0.014)	0.002	0.029** (0.011)	0.003
N	270		753		383		573	
ln(L)	-44.23		-176.9		-76.90		-121.6	
count R ²	0.937		0.894		0.909		0.904	
adj count R ²	0.871		0.775		0.794		0.808	

Robust standard errors in parentheses.

** p<0.01, * p<0.05, + p<0.1. Except for HOME, the marginal effects (*m.e.*) are calculated evaluating equation (5) at each observation, and averaging over the observations. Because the predicted probabilities lie in the range [0,1], the marginal effects measure change in this range.

which both teams had a lot at stake was 9.2%, about twice as large as the home advantage when no more than one of the teams had more than 85 points, but again, we cannot deduce the specific cause for this difference. It is consistent with referee bias being larger when more is at stake, but it is also consistent with the idea that the home team plays more aggressively when more is at stake *and* they are playing in front of their fans.

6. CONCLUSION

In this paper we have analyzed the home advantage in the NHL from two different perspectives. Using season-level data we find no systematic variation in the home advantage across teams. The advantage varies over time, but there is little evidence to suggest that it is anything but random. In order to uncover the sources of the home advantage, it is necessary to use game-level data. We use six seasons of game-level data to study the oft cited sources of better rest, rules advantages, crowd effects and referee bias. Our results show that more rest before games has no effect on the win-loss probability for the home team. We find good evidence that the rules advantage for the home team regarding stick placement during face-offs contributes to the home advantage, but we find no evidence that having the last line change opportunity benefits the home team. We found no evidence of a crowd-size effect on the win-loss probability. As for referee bias, we found some evidence that is consistent with referee bias, but we cannot argue that it is caused by referee bias. This was evident in our finding that home teams are no more likely to win an overtime game that goes to a shootout, but they are more likely to win a game that ends in overtime compared to a game that ends in regulation with a one goal differential. In order to flush out this effect, we would require more micro data, such as period by period data. Information about the awarding of penalty minutes during overtime, could be used to study referee bias. Our finding that the home team has a larger advantage when there is more at stake with regard to the playoffs is consistent with a number of hypotheses, including referee bias, but also consistent with the psychological effects of crowd support. What is needed is a clever identification strategy akin to the empty stadium natural experiment that was studied in soccer by Pettersson-Lidbom and Priks (2010).

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